

DS302 Data Science Methodology

Super Store Sales Performance Analysis

Final Project Report Power BI Retail Sales Dashboard

Project Team

Mohamed El-Sayed Aly 202200655
Ahmed Mahmoud 202200460

Supervised by

Dr. Soha

Eng. Hamdy Saied

Spring 2025 – 2026

Section 1 — Project Introduction

1. Project Overview

This report documents the complete development of a Business Intelligence (BI) solution built using Microsoft Power BI. The project demonstrates the full BI development lifecycle from raw data ingestion and transformation, through data modeling and DAX calculations, to the final interactive dashboard with professional Figma-designed visualizations.

1.1 Objectives

- Build a fully functional, interactive Power BI dashboard for a retail sales organization
- Apply data cleaning and transformation techniques using Power Query
- Implement a Star Schema data model with proper relationships
- Create +13 DAX measures covering revenue, profit, growth, and time intelligence
- Design 4 professional report pages with dark-themed Figma backgrounds
- Demonstrate complete BI development lifecycle from raw data to published insights

1.2 Problem Statement

A fictional global retail company Superstore operates across multiple regions in the United States with thousands of transactions across three product categories: Technology, Furniture, and Office Supplies. The management lacks a centralized, visual way to monitor sales performance, customer behavior, product profitability, and regional trends. This project delivers a complete BI solution that addresses these business needs through interactive dashboards.

1.3 Dataset Description

The project uses the Superstore Sales Dataset sourced from Kaggle (by Vivek Chowdhury). The dataset contains transactional data for a fictional global retail company with customers across multiple US regions.

Property	Details
Source	Kaggle Superstore Sales Dataset (vivek468)
File Name	Sample - Superstore.csv
Total Rows	9,994 transactions
Total Columns	21 columns
Date Range	2014 – 2017
Geography	United States — 4 Regions, 49 States
Categories	Technology, Furniture, Office Supplies

1.4 Dataset Columns

The raw dataset contains the following 21 columns:

Order ID	Order Date	Ship Date	Ship Mode
Customer ID	Customer Name	Segment	Country
City	State	Postal Code	Region
Product ID	Product Name	Category	Sub-Category
Sales (Revenue)	Quantity	Discount	Profit
Row ID	+ Profit Status (added)		

Section 2 — Data Cleaning & Transformation

2. Power Query — Data Cleaning

After loading the CSV file into Power BI, the Transform Data option was selected to enter the Power Query Editor. Six transformation steps were applied to ensure data quality and consistency before modeling.

Step 1 — Remove Duplicates

Duplicate rows were removed using the Order ID column as the unique key. This ensures each transaction appears only once in the dataset, preventing double-counting in calculations.

Step 2 — Handle Null Values

Null values in the Revenue (Sales) and Profit columns were replaced with 0. This prevents calculation errors in DAX measures and ensures financial aggregations return accurate results. For number columns, the keyword 'null' was used in the Value To Find field.

Step 3 — Enforce Data Types

Each column was assigned the correct data type to ensure accurate calculations and proper visual rendering in Power BI.

Column	Data Type	Reason
Order Date, Ship Date	Date	Enable time intelligence in DAX
Revenue, Profit, Discount	Decimal Number	Allow precise financial calculations
Quantity	Whole Number	Quantities are always integers
Order ID, Customer ID, Product ID	Text	IDs should not be aggregated

Step 4 — Add Profit Status Column

A custom calculated column called Profit Status was added to classify each transaction based on its profit value. This column enables quick analysis of business performance at the order level.

Condition	Result	Business Meaning
Profit > 0	Profitable	The transaction generated a positive return
Profit = 0	Break-Even	No gain or loss on this transaction
Profit < 0	Loss	The transaction resulted in a financial loss

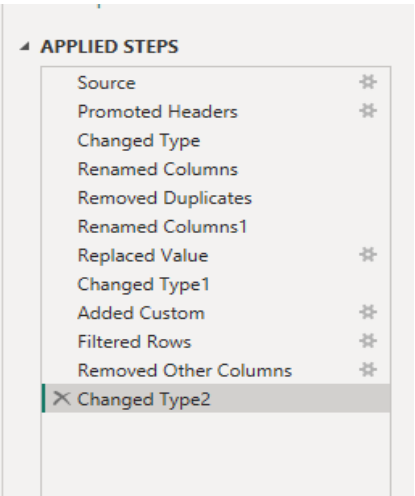
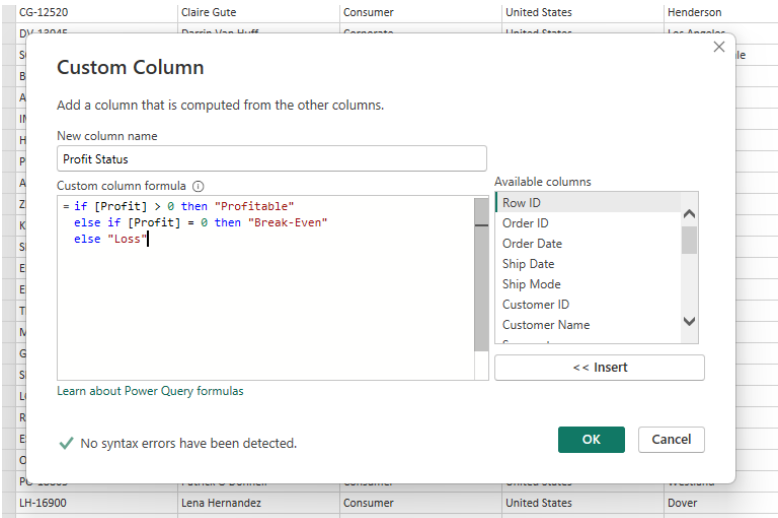
Step 5 — Filter Test Rows

Rows where Order ID is null were removed. These represent corrupted or test records that were accidentally included in the raw data export. Removing them ensures data integrity throughout all calculations and visualizations.

Step 6 — Rename Columns

Column names were standardized for consistency across all tables and DAX measures.

Original Name	New Name
Sales	Revenue
Cust ID	CustomerID



Section 3 — Data Modeling (Star Schema)

3. Star Schema Data Model

After completing the Power Query transformation, a Star Schema was implemented to organize the data into a proper dimensional model. The Star Schema separates data into one central Fact table and four surrounding Dimension tables, connected by Many-to-One relationships.

3.1 Schema Overview

The Star Schema consists of 5 tables in total:

- **Fact Table:** 1 Fact Table contains all **transactional measures**
- **Dimension Tables:** 4 Dimension Tables contain **descriptive attributes**

3.2 Entity Relationship Diagram (ERD)

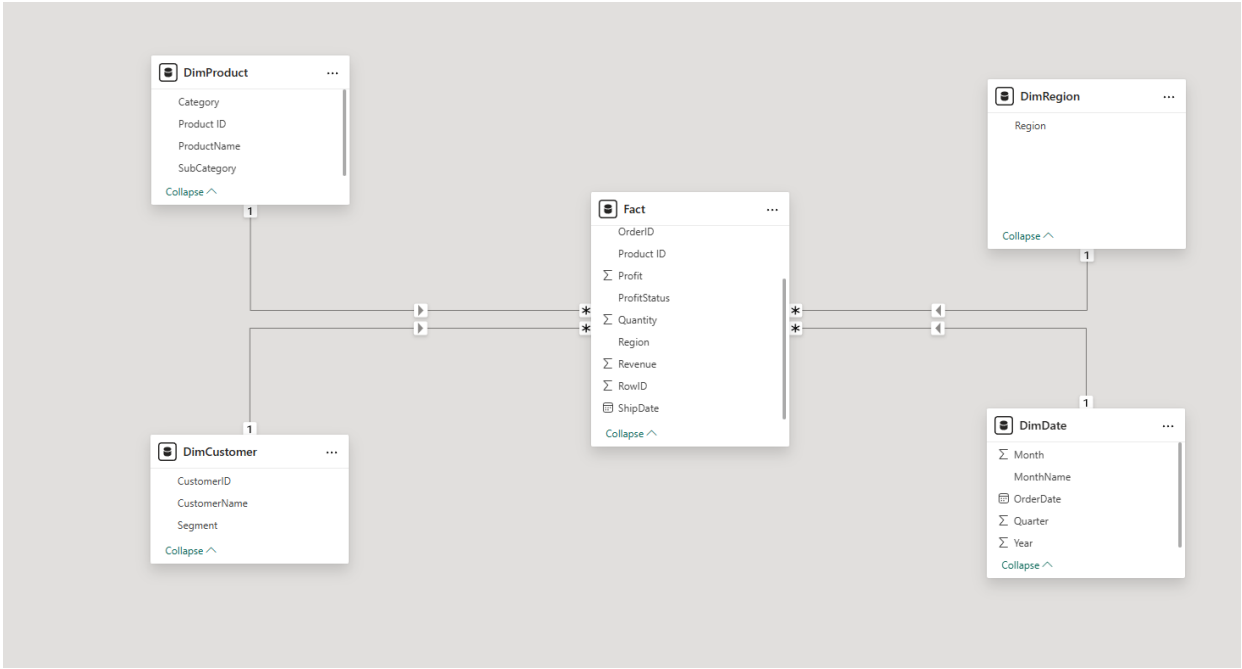
FACT TABLE — Fact_Sales					
OrderID (FK) CustomerID (FK) ProductID (FK) OrderDate (FK) Region (FK)					
Revenue Profit Quantity Discount ProfitStatus RowID					

DimCustomer	DimProduct	DimDate	DimRegion
CustomerID (PK) Customer Name Segment	ProductID (PK) Product Name Category Sub-Category	OrderDate (PK) Year Month Month Name Quarter	Region (PK)

3.3 Relationships

All relationships are Many-to-One from the Fact table to each Dimension table:

From (Fact Table)	Relationship	To (Dimension Table)	Key Column
Fact[CustomerID]	Many → One	DimCustomer[CustomerID]	CustomerID
Fact[ProductID]	Many → One	DimProduct[ProductID]	ProductID
Fact[OrderDate]	Many → One	DimDate[OrderDate]	OrderDate
Fact[Region]	Many → One	DimRegion[Region]	Region



Section 4 — DAX Measures

4. DAX Measures Documentation

A total of +13 DAX measures were created to support all report pages. These measures are organized into four categories: Basic Aggregations, Customer Intelligence, Financial Analysis, and Time Intelligence.

4.1 Basic Aggregations

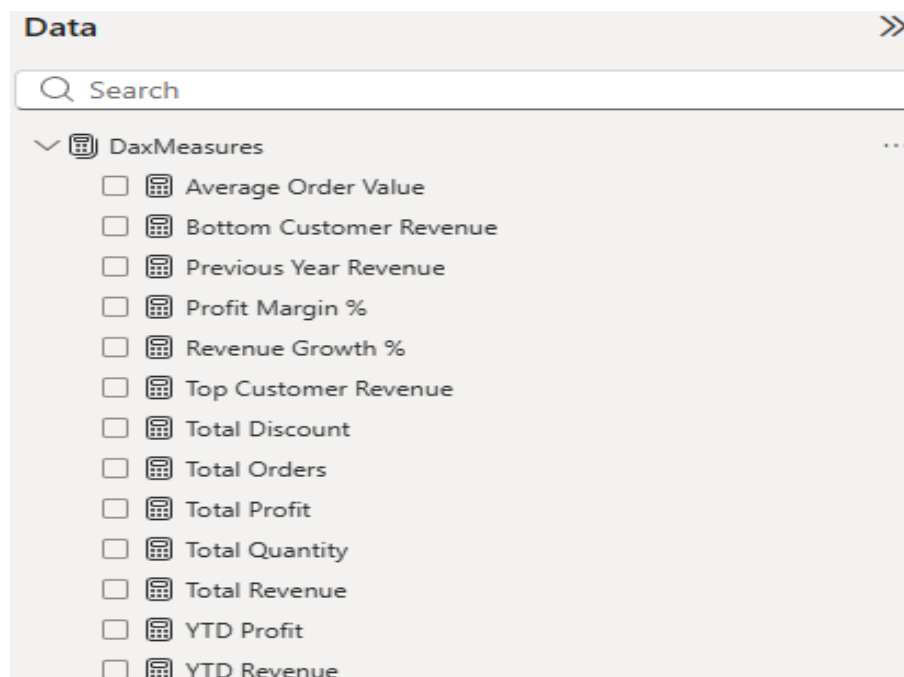
Measure Name	DAX Formula	Description	Sample Output
Total Revenue	<code>SUM(Fact[Revenue])</code>	Sum of all revenue across all transactions	1.10M
Total Profit	<code>SUM(Fact[Profit])</code>	Sum of all profit values	132.52K
Total Quantity	<code>SUM(Fact[Quantity])</code>	Total units sold across all orders	38K
Total Orders	<code>DISTINCTCOUNT(Fact[OrderID])</code>	Count of unique order transactions	5,009
Total Discount	<code>SUM(Fact[Discount])</code>	Sum of all discount values applied	786.45

4.2 Customer & Financial Intelligence

Measure Name	DAX Formula	Description	Sample Output
Average Order Value	<code>DIVIDE([Total Revenue], [Total Orders])</code>	Average revenue generated per order	219.58
Profit Margin %	<code>DIVIDE([Total Profit], [Total Revenue]) * 100</code>	Percentage of revenue retained as profit	12.05%
Top Customer Revenue	<code>MAXX(VALUES(DimCustomer[Customer Name]), [Total Revenue])</code>	Revenue of the single highest-spending customer	12.12K
Bottom Customer Revenue	<code>MINX(VALUES(DimCustomer[Customer Name]), [Total Revenue])</code>	Revenue of the single lowest-spending customer	4.83

4.3 Time Intelligence Measures

Measure Name	DAX Formula	Description	Sample Output
YTD Revenue	TOTALYTD([Total Revenue], DimDate[OrderDate])	Cumulative revenue from Jan 1 to current date	357.26K
YTD Profit	TOTALYTD([Total Profit], DimDate[OrderDate])	Cumulative profit from Jan 1 to current date	44.71K
Previous Year Revenue	CALCULATE([Total Revenue], SAMEPERIODLASTYEAR(DimDate[OrderDate]))	Total revenue for the same period last year	609.21K
Revenue Growth %	DIVIDE([Total Revenue] - [Previous Year Revenue], [Previous Year Revenue]) * 100	Year-over-year revenue growth percentage	48.11%



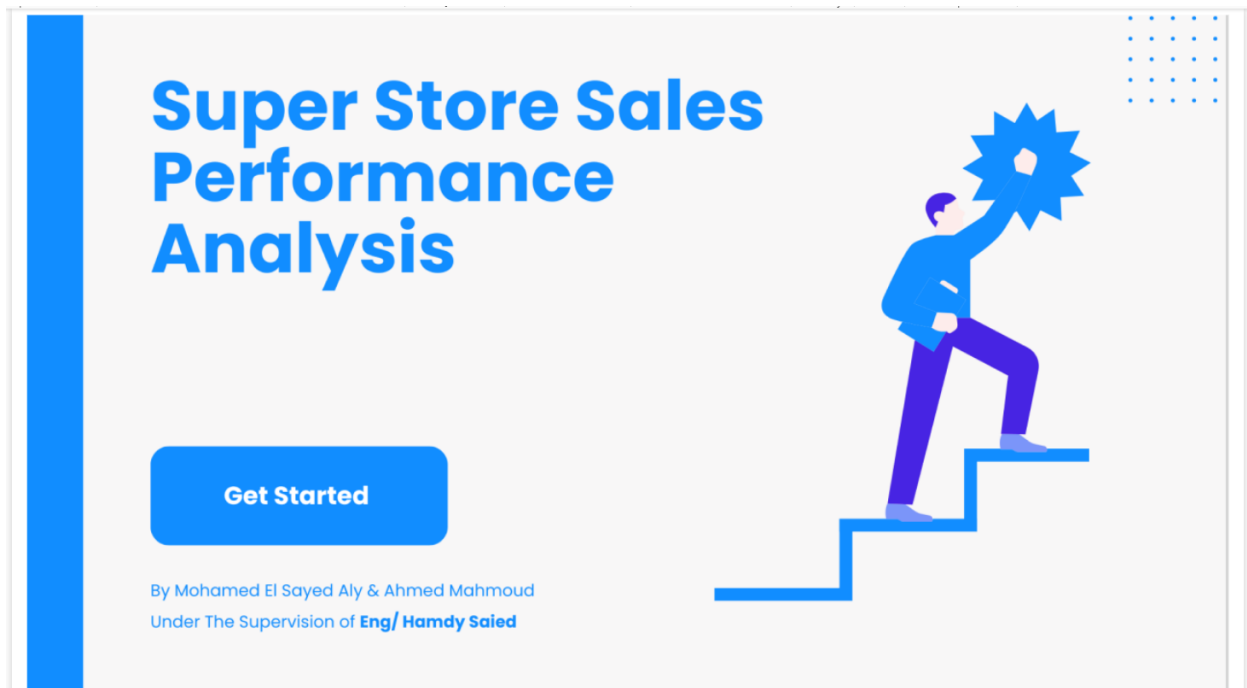
Section 5 — Power BI Report Pages

5. Report Pages Design

The Power BI report consists of 5 pages: a Home Screen and 4 analytical pages. Each page features a professionally designed dark background created in Figma and imported as a PNG. All pages use slicers on the left panel for interactivity.

Home Screen

A dedicated landing page was designed featuring the project title, team names, supervisor information, and a 'Get Started' navigation button. This page serves as the entry point to the report and demonstrates professional BI report design.



Page 1 — Revenue Overview

This page provides a high-level overview of overall sales performance. Theme color: Blue (#58A6FF).

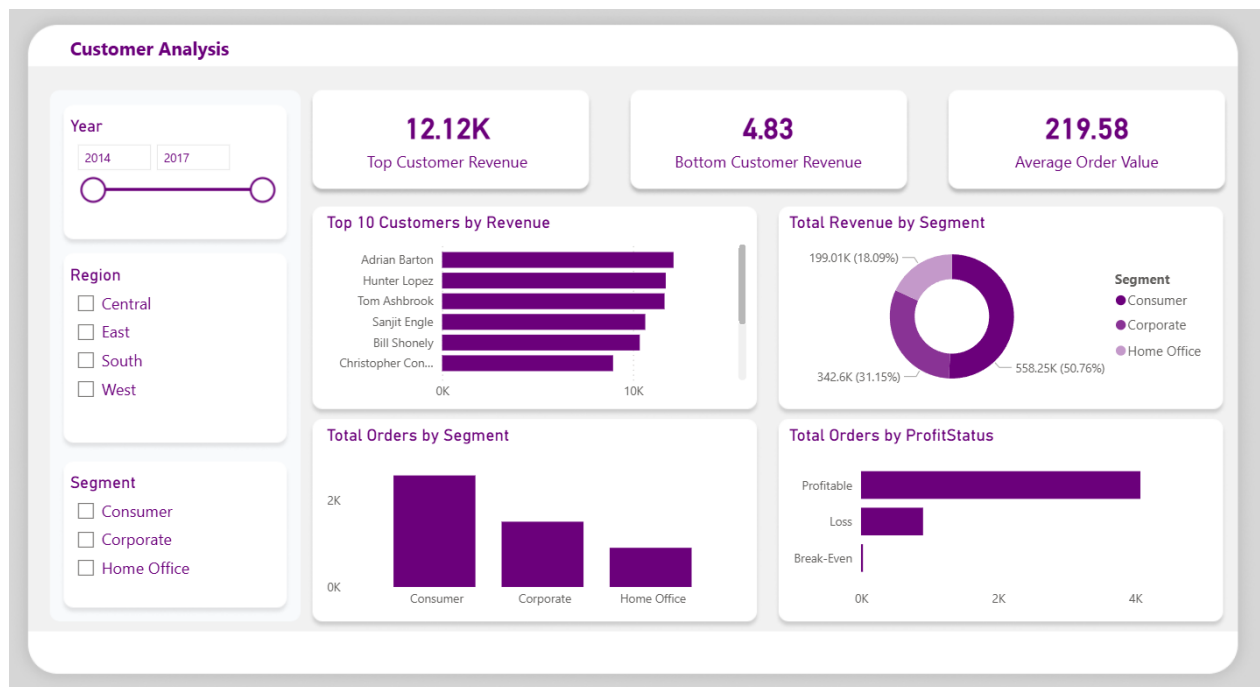
Visual	Type	Fields Used
KPI Card 1	Card	Total Revenue
KPI Card 2	Card	Total Orders
KPI Card 3	Card	Total Quantity
Revenue over Years	Line Chart	X: Order Date Y: Total Revenue
Revenue by Category	Bar Chart	Y: Category X: Total Revenue
Monthly YTD Revenue	Column Chart	X: Month Y: YTD Revenue
Revenue by Segment	Donut Chart	Legend: Segment Values: Total Revenue
Slicers (Left Panel)	Slicer	Year Region Category



Page 2 — Customer Analysis

This page analyzes customer behavior, spending patterns, and segment distribution. Theme color: Purple (#BC8CFF).

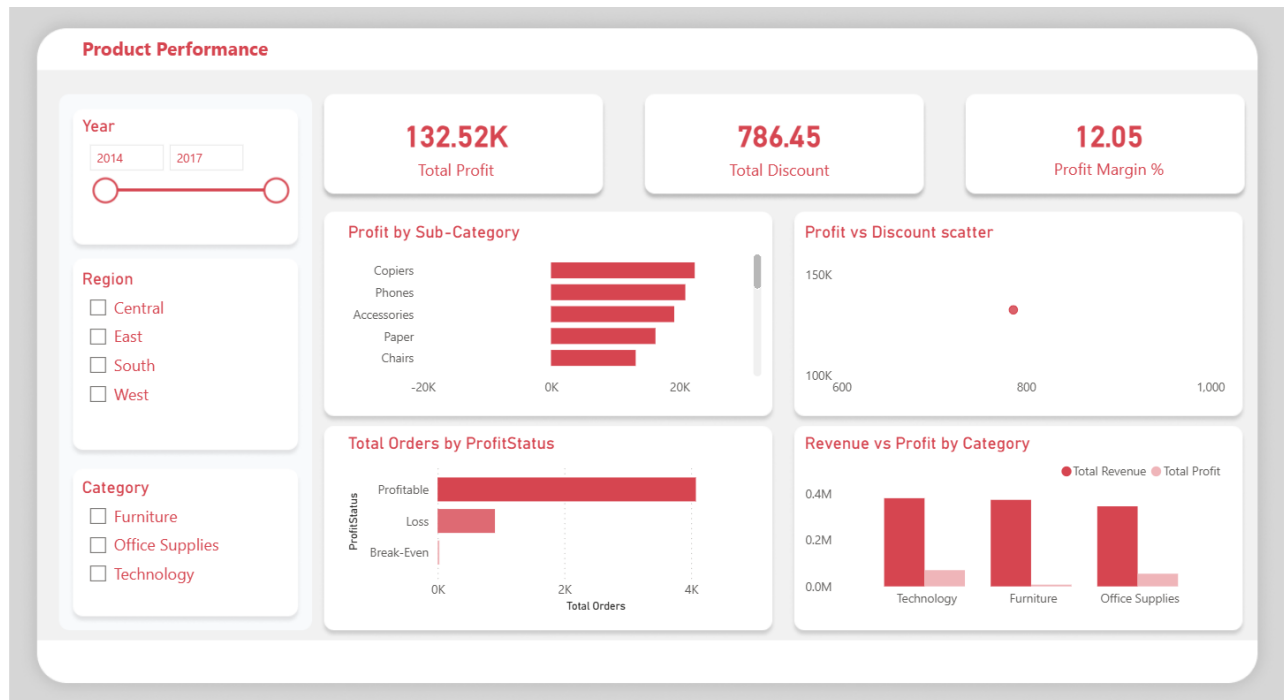
Visual	Type	Fields Used
KPI Card 1	Card	Top Customer Revenue
KPI Card 2	Card	Bottom Customer Revenue
KPI Card 3	Card	Average Order Value
Top 10 Customers	Bar Chart	Y: Customer Name (Top N=10) X: Total Revenue
Revenue by Segment	Donut Chart	Legend: Segment Values: Total Revenue
Orders by Segment	Column Chart	X: Segment Y: Total Orders
Orders by Profit Status	Bar Chart	Y: ProfitStatus X: Total Orders
Slicers (Left Panel)	Slicer	Year Region Segment



Page 3 — Product Performance

This page evaluates profitability and performance across product categories and sub-categories. Theme color: Red (#FF4444).

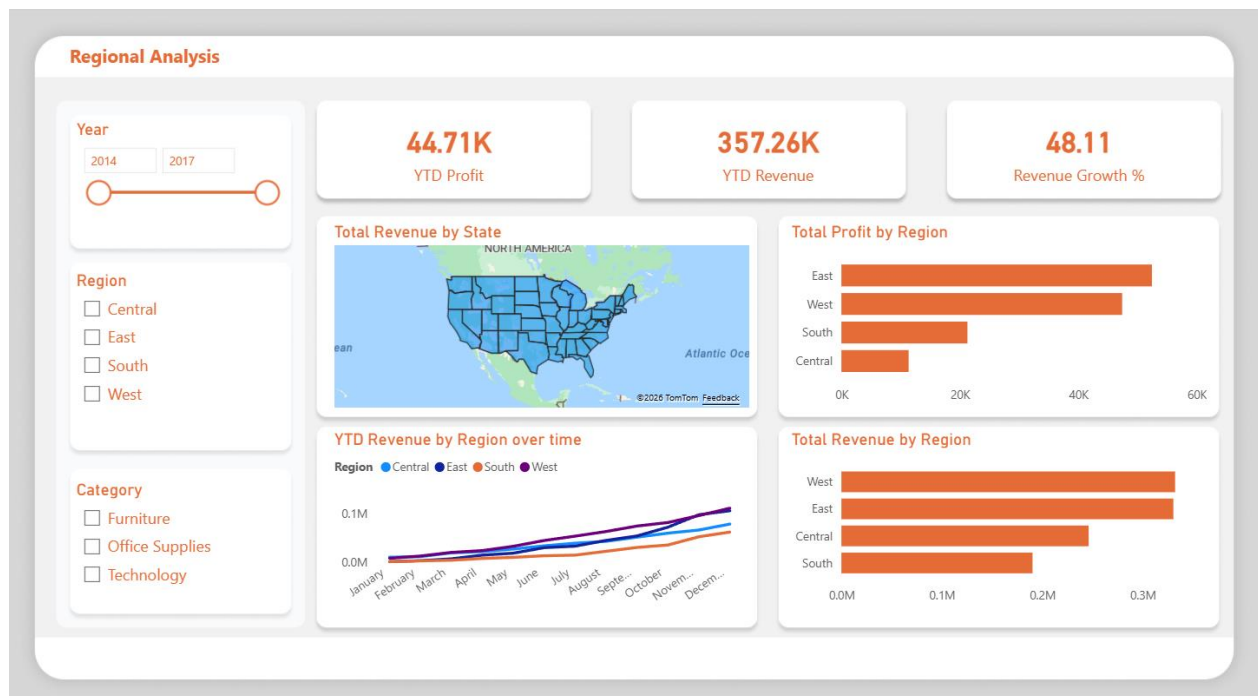
Visual	Type	Fields Used
KPI Card 1	Card	Total Profit
KPI Card 2	Card	Total Discount
KPI Card 3	Card	Profit Margin %
Profit by Sub-Category	Bar Chart	Y: Sub-Category X: Total Profit
Profit vs Discount Scatter	Scatter Chart	X: Total Discount Y: Total Profit Details: Sub-Category
Orders by Profit Status	Donut Chart	Legend: ProfitStatus Values: Total Orders
Revenue vs Profit by Category	Column Chart	X: Category Y: Total Revenue & Total Profit
Slicers (Left Panel)	Slicer	Year Category Region



Page 4 — Regional Analysis

This page focuses on geographic and regional sales performance with time-based trends.
Theme color: Orange (#FF7B35).

Visual	Type	Fields Used
KPI Card 1	Card	YTD Profit
KPI Card 2	Card	YTD Revenue
KPI Card 3	Card	Revenue Growth %
Revenue by State	Filled Map	Location: State Color: Total Revenue
Profit by Region	Bar Chart	Y: Region X: Total Profit
YTD Revenue by Region	Line Chart	X: Month Y: YTD Revenue Legend: Region
Revenue by Region	Bar Chart	Y: Region X: Total Revenue
Slicers (Left Panel)	Slicer	Year Region Category

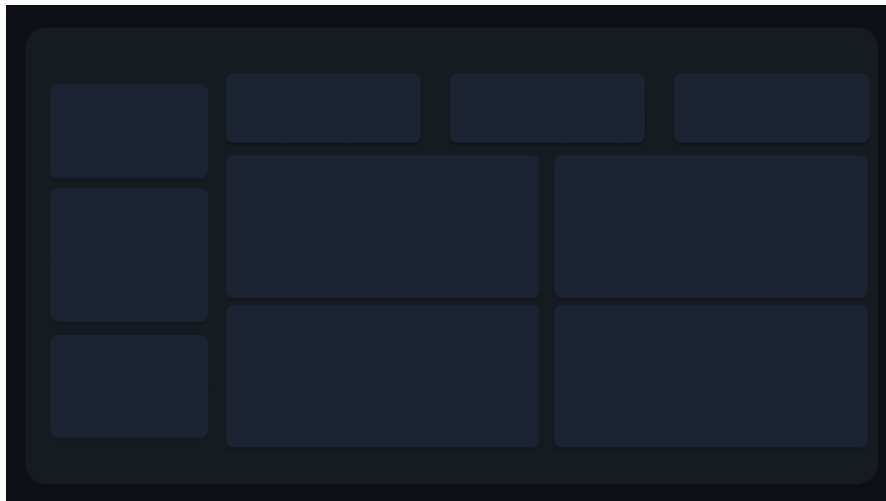
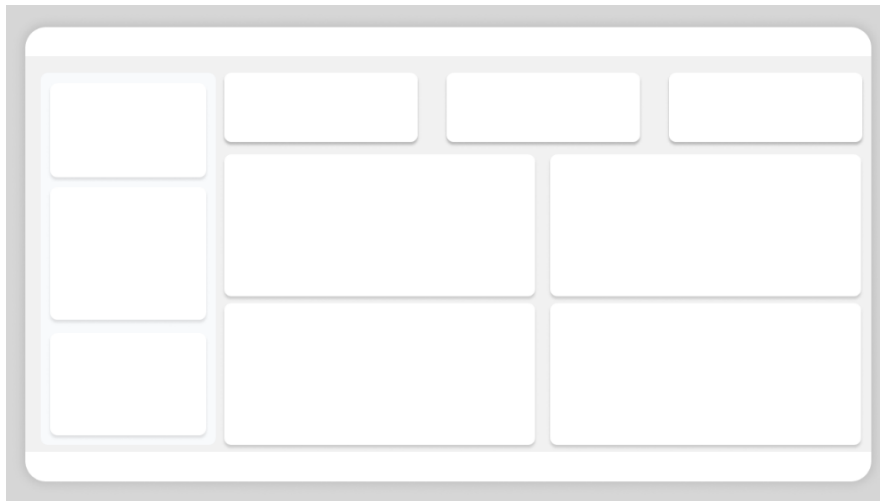


Section 6 — Figma Designs

6. Design & Visual Theme

The Power BI report was designed in two stages. First, a light version of the dashboard was built directly in Power BI to verify that all visuals, fields, and DAX measures were correctly configured and displaying accurate data. Once the data layer was confirmed, a professional dark-themed background was designed for each of the four report pages using Figma, a professional UI design tool.

Figma Link: <https://www.figma.com/design/8X1iivNMQ4clkXGVwsqgv6/Project-Dashboard?node-id=0-1&t=KFL7OOW6k5Mm0fWI-1>



Section 7 — Key Business Findings

7. Key Business Findings

The following insights were derived from the completed Power BI dashboard analysis of the Superstore dataset:

7.1 Revenue Performance

- Total Revenue across all years (2014-2017) reached 1.10 million dollars
- Revenue showed consistent year-over-year growth of 48.11% in the final year
- Technology is the highest revenue category, followed by Furniture and Office Supplies
- The Consumer segment dominates with 50.76% of total revenue

7.2 Profitability Insights

- Total Profit is 132.52K, representing a Profit Margin of 12.05%
- Several Sub-Categories such as Tables and Bookcases show negative profit — indicating loss-making products
- 81.37% of all orders are profitable, 18.07% result in a loss, and only 0.56% break even
- High discount rates are correlated with lower or negative profit in the scatter analysis

7.3 Regional Performance

- The East region generates the highest total profit followed by the West region
- The Central region shows the lowest profit margin, suggesting operational inefficiencies
- The West region leads in total revenue volume
- YTD Revenue shows consistent growth across all 4 regions throughout each year

7.4 Customer Analysis

- Adrian Barton is the top customer with approximately 12.12K in revenue
- Average Order Value stands at 219.58 dollars per transaction
- Corporate segment shows higher average order values compared to Consumer
- Home Office segment represents the smallest portion at 18.09% of revenue

Section 8 — Conclusion

8. Conclusion

This project successfully delivered a complete, professional Business Intelligence solution for the Superstore retail dataset. The following deliverables were completed:

#	Deliverable	Status
1	Data loading from Kaggle Superstore CSV	Completed
2	6 Power Query transformation steps applied	Completed
3	Star Schema with 1 Fact + 4 Dimension tables	Completed
4	+13 DAX measures covering all analytical areas	Completed
5	4 report pages with Figma backgrounds Light and Dark	Completed
6	Professional Home Screen with navigation button	Completed
7	Interactive slicers on all pages (Year, Region, Category)	Completed
8	Project documentation report (.docx)	Completed
9	Project Published on Power BI Services (Bonus Point)	Completed

The project demonstrates a solid understanding of the complete BI development lifecycle, from raw data ingestion through to a polished, interactive dashboard. The use of Star Schema modeling, DAX time intelligence measures, and Figma-designed backgrounds reflects professional industry standards.

Power BI Services Link:

<https://app.powerbi.com/groups/me/reports/bd743354-a437-4e25-be45-53503ef4de2e/d03c668243ac178ec0c3?experience=power-bi>